Evan Litzer CS230

1A. $\exists$ **x (C(x)** $\wedge$ **R(x))** $\wedge$ $\forall$ **y (C(y)** $\rightarrow \neg$ **D(x, y))**
$\exists$ C(x): There exists someone in the class.
$\wedge$ R(x): And they have read the material.
$\wedge$ $\forall$ y (C(y) $\rightarrow$ D(x, y)): And if all the students are also in the class, then they have not discussed it with some student.
1B. $\exists$ **x** $\exists$ **y (C(x)** $\wedge$ **C(y))** $\wedge$ $\forall$ **z (C(z)** $\rightarrow$ **(D(y, z)** $\vee$ **D(x, z)))**
$\exists$ x $\exists$ y (C(x) $\wedge$ C(y)) : Some two students who are both in your class.
$\wedge$ $\forall$ z (C(z) : And if all of the students (z) are also in the class.
$\rightarrow$ (D(y, z) $\vee$ D(x, z))) : Then all of the students have discussed the material with the two students.

2A:
Predicates: P(x) = x has visited Paris | C(x) = x is in class | A(x) = x has visited Arc de Triomphe
Premises: Mary is a student in class. Everyone who goes to Paris visits Arc de Triomphe.
Conclusion: Someone in the class has visited Arc de Triomphe.

$\exists$ x (C(x) $\wedge$ P(x))	P1
$\forall$ x (P(x) $\rightarrow$ A(x))	P2
C(a) $\wedge$ P(a) for some a	Existential Instantiation (1)
P(a) $\rightarrow$ A(a) for all a	Universal Instantiation (2)
P(a) for all a	Simplification (4)
A(a) for all a	Modus Ponens (4, 5)
C(a) for some a	Simplification (3)
C(a) $\wedge$ A(a) for some a	Conjunction (6, 7)
$\exists$ x (C(x) $\wedge$ A(x))	Existential Generalization (8)

2B.
Predicates: C(x) = College towns | M(x) = Midwest towns | F(x) = Fun places to live.
Premises: College towns in midwest. All college towns fun places to live.
Conclusion: There is a town in the midwest that is fun to live in.

$\exists$x (C(x) $\wedge$ M(x))	P1
$\forall$ x (C(x) $\rightarrow$ F(x))	P2
C(a) $\wedge$ M(a) for some a	Existential Instantiation (1)
C(a) $\rightarrow$ F(a) for all a	Universal instantiation (2)
F(a) for all a	Modus Ponens (3, 4)

M(a) for some a	Simplification (2)
F(a) ∧ M(a) for some a	Conjunction (5, 6)
∃x (F(x) ∧ M(x))	Existential generalization (7)

2C:

Predicates: B(x) = x is bear | S(x) = x is good swimmer | C(x) = x can catch fish | H(x) = x goes hungry

Premises: All bears are good swimmers. If you can catch fish, you will not go hungry. If you can't catch fish, you are not a good swimmer.

Conclusion: Therefore, no bears go hungry.

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∀ x (B(x) → S(x))	P1
∀ x (C(x) → ¬H(x))	P2
∀ x (¬ C(x) → ¬ S(x))	P3
B(a) → S(a) for all a	Universal Instantiation (1)
C(a) → ¬H(a) for all a	Universal Instantiation (2)
¬B(a) → ¬S(a) for all a	Universal Instantiation (3)
S(a) → B(a) for all a	Contrapositive (6)
B(a) → C(a) for all a	Hypo Syllogism (4, 7)
B(a) → ¬ H(a) for all a	Hypo Syllogism (5, 8)
∀x(B(x) → ¬H(x))	Universal Generalization (9)

3.

Predicates: P(x) = x is odd | Q(x) = x is even

Domain: Set of integers

∃x(P(x)∧Q(x))	This statement proposes that there can be some integer x that is both odd and even, which is impossible according to the rules of the integer. This statement will always result in being false.
∃xP(x) ∧ ∃xQ(x)	This statement proposes that there is some integer x that can be odd and some integer x that can be even. This statement will always result in being true.

4. Universe: N = {0, 1, 2, 3…}
Given Predicates: A(n, m, k) = n + m = k | M(n, m, k) = n * m = k
New Predicates: Zero(n) = A(n, n, n) = True when n=0, false otherwise | Greater(m, n)
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4A: Equal(m, n) = ¬Greater(n, m) ⋀ ¬ Greater(m, n)
¬Greater(n, m): N is not greater than M
¬ Greater(m, n): M is not greater than N

4B: One(n) = M(n, n, n) ⋀ ¬Zero(n)
- M(n, n, n) with n = 1 will always compute to 1.
- Since 0 is the only other number that will equal itself when multiplied by itself, ¬Zero(n) must also be included.

4C: Two(n) = One(m) ⋀ A(m, m, n)
- One(m) locks in m as a 1 value..
- A(m, m, n) locks in n as a 2 value as 1 + 1 = 2.